

Yu Li

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Education

George Washington University <i>Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0</i> Advisor: Prof. Tian Lan ✉ ; Work with: Prof. Zhengling Qi ✉	Washington, D.C. 2025.08 – 2029.05 (expected)
Wuhan University, Hongyi Honor College <i>B.Eng. in Microelectronics Science and Engineering, GPA: 3.87/4.0</i>	Wuhan, China 2021.09 – 2025.06

Experience

ByteDance, Tiktok <i>Research Scientist Intern, Recommendation Core</i>	San Jose, CA 2026.05 – 2026.08
Owned the ranking stage of TikTok video recommendation, working on 3B Large Recommendation Model with MLA, time-aware RoPE, and mixed-query optimization.	

Research Highlights

My research develops principled reinforcement learning formulations for **LLM Reasoning**, spanning token-level credit assignment, preference optimization, and context conditioning [\[NeurIPS'26\]](#) [\[NeurIPS'26\]](#)[\[P2\]](#)[\[P3\]](#), and extends these formulations to **Agentic RL** through hierarchical skill acquisition and multi-turn policy optimization [\[P1\]](#)[\[ACL'26\]](#)[\[INFOCOM'26\]](#). On **Efficient AI**, my work focuses on parameter-efficient fine-tuning and long-context KV cache management [\[COLM'26\]](#)[\[AAAI'26\]](#). Earlier work on **Generative Models** covers diffusion-based personalized content-style synthesis [\[CVPR'26\]](#).

Preprints & Under Review

P=Preprint

- [\[P1\]](#) **ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning** [✉](#)
[Yu Li](#), Rui Miao, Zhengling Qi, Tian Lan
- [\[P2\]](#) **Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO** [✉](#)
[Yu Li](#), Tian Lan, Zhengling Qi
- [\[P3\]](#) **Hindsight Self-Distillation: Demonstrating to the Teacher for On-Policy LLM Reasoning** [✉](#)
[Yu Li](#), Shu Hong, Tian Lan
Under review. ARR May 2026 OA: 3.33/5

Selected Publications [\[Full List\]](#)

†=Equal Contribution

I have published **9** papers at top-tier conferences, with **7** of them as **first or co-first author**.

- [\[NeurIPS'26\]](#) **OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning**
[Yu Li](#), Rui Miao, Tian Lan, Zhengling Qi
- [\[NeurIPS'26\]](#) **INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization**
[Yu Li](#), Tian Lan, Zhengling Qi

[COLM'26] MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference

Yu Li, Binxu Li, Tian Lan

[ACL'26 Findings] Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization

Yu Li, Sizhe Tang, Tian Lan

[CVPR'26] CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion

Yu Li, Yujun Cai, Chi Zhang

[ICASSP'26 Oral] KG-SAM: Encoding Structural Constraints into Segment Anything Models via Probabilistic Graphical Models

Yu Li, Da Chang, Xi Xiao

[INFOCOM'26] ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense

Yu Li[†], Sizhe Tang[†], Rongqian Chen, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D. Bastian, Tian Lan

[ACM MM'26] Prompted Information Bottlenecks: Rethinking Layer-Wise Information Allocation in Frozen Vision Transformers

Yuqi Li, Xi Xiao, Yunbei Zhang, Lin Zhao, Yu Li, Aiden Zhao, Tianyang Wang, Hao Xu, Yingli Tian

[AAAI'26] Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT

Da Chang, Peng Xue, Yu Li, Yongxiang Liu, Pengxiang Xu, Shixun Zhang

Selected Awards & Honors

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- ICML 2026 Gold Reviewer, top reviewers recognition. 2026
 - Innova International Exchange Scholarship, 6 recipients university-wide, ¥70,000. 2024
 - Innova Excellence Scholarship, Top 3%, ¥10,000, twice. 2023, 2024
 - First-Class Scholarship, Top 5%, ¥3,000, three times. 2022, 2023, 2024

Academic Services

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- Conference Reviewer: 2027: AAAI. 2026: NeurIPS, ICML, ICLR, COLM, AAAI, ACM MM.
 - Journal Reviewer: TPAMI, TMLR, ToN, Neurocomputing.